In-Context Next-Word Prediction as a Mechanism for Generating Focus Alternatives

Human communication requires not just understanding the literal meaning of an utterance but also inferring what *could* have been said instead (Grice, 1975). Of course, there are too many things a speaker could have said to consider them all. Given the context, listeners must consider only the alternatives they take to be relevant. For example, in order to compute the meaning of the utterance, "The chef baked only cookies," a listener must generate and rule out alternatives for the constituent within the scope of *only* (e.g. cake, muffins, pie). Prior work (Rooth, 1992; Fox & Katzir, 2011; Degen, 2023) often assumes the existence of contextually relevant alternatives, but there exist no formal or computational models of how these alternatives are generated in the first place. We hypothesize that when a word is within a focus-marked constituent, people consider other words that would have been high probability completions for the prior context (e.g. "The chef baked __") as alternatives. We test the hypothesis that in-context next-word prediction, as modeled by large language models (LLMs), can serve as a mechanism for focus alternative generation.

Methods: We use LLMs as a computational model of human next-word prediction for two reasons. First, methodologically, human cloze probability judgments are difficult to obtain for low probability words, and these could be among the alternatives people consider. LLMs allow us to get a full next-word probability distribution across an entire vocabulary. Second, while the mechanisms driving human next-word prediction remain unknown (Ryskin & Nieuwland, 2023), LLMs provide at least a mathematically (if not cognitively) transparent model of how next-word distributions are calculated from context.

To validate using LLMs as a proxy for human next-word predictions, we obtain Spearman correlations between next-word predictions from five LLMs (BERT, GPT2, Qwen, DeepSeek, and LLaMA) and two cloze probability datasets (a novel dataset and Peelle et al., 2020) [Figure 1]. To determine a good model of focus alternatives, we designed experiments measuring participants' judgments on focus-sensitive inferences [Figure 2]. Inspired by formal semantic theories, we modeled four candidate structures for reasoning about focus alternatives: Set-based (words above a probability threshold of being the next word are negated), Ordering-based (words ranked more probable than what was said are negated), and their conjunction and disjunction [Figure 3 shows how the alternative structures make different predictions]. We compare the predictions the alternative structures make given the different next-word predictions from both BERT and BERT's static embedding space, to the experimental data (using log-likelihoods).

Results: The correlations between the next-word predictions of the LLMs and the two cloze probability datasets were moderate and significant (all $p < 0.001$), with Qwen's predictions being the most correlated and GPT-2 the least [Table 1]. Using BERT as a test case, we found that dynamic next-word prediction outperforms static embeddings in modeling human focus alternative judgments [Table 2]. This suggests that context-sensitive next-word prediction models alternative generation better than nearest-neighbors in a static semantic space. The Ordering structure provided the best fit to the human data, suggesting that alternatives are on average negated *in proportion* to their predictability in-context. Having established that in-context next-word prediction is a plausible mechanism for generating focus alternatives (via BERT) and that Qwen is the better model of human next-word prediction (via correlations on cloze tasks), we are currently testing whether Qwen outperforms BERT in focus alternative generation, in order to test the hypothesis that better models of next-word prediction are also better models of focus alternative generation.

Conclusion: We provide computational evidence that focus alternatives can be generated via in-context next-word prediction. Judgments about which alternatives to X are negated by saying "only X" are well predicted by LLMs' probability of predicting those words in place of X in the same context, suggesting that listeners' predictive processing feeds into the ad hoc alternatives they consider.

Figures and Tables

Figure 1: Examples of the cloze task from our novel dataset and Peelle et al. (2020), including example next words from participants.

	Context / Sentence	Example next words from participants
Our novel dataset (10 contexts)	You and your friend Sam go for a long walk together. After the walk you go back to Sam's house. You say to Sam, "I'm thirsty." Sam opens the fridge and responds, "Sure, I have ___."	water, soda, juice, milk, beer, tea, lemonade
Peelle et al. (2020) (~3,000 sentences)	He hated bees and feared encountering a ___.	hive, swarm, bee, nest, wasp

Figure 2: Example of an experimental trial. All trials had the same template: context, a request that defines the question under discussion (QUD), and a response to the QUD. Following each story, participants answered a multiple choice question.

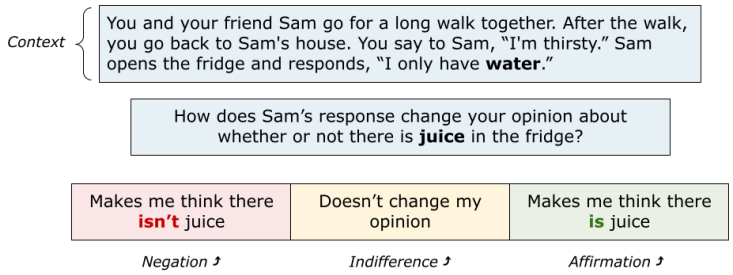

Figure 3: The human judgement prediction of different focus alternative structures given a context.

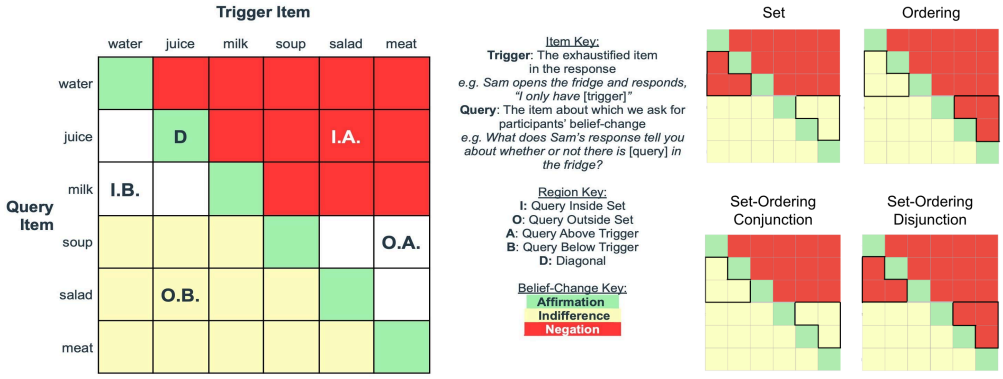

Table 1: Spearman correlation results between five LLMs (BERT, GPT2, Qwen, DeepSeek, and LLaMA) and two human cloze probability datasets (a novel dataset and Peelle et al., 2020).

LLM	Our novel dataset: Correlation across all contexts	Peelle et al. (2020): Correlation across all sentences
Qwen 2-7B	0.549 ($p < 0.001$)	0.490 ($p < 0.001$)
LLaMA 3.2-3B	0.500 ($p < 0.001$)	0.477 ($p < 0.001$)
DeepSeek Coder 1.3B	0.435 ($p < 0.001$)	0.316 ($p < 0.001$)
BERT	0.419 ($p < 0.001$)	0.365 ($p < 0.001$)
GPT-2	0.389 ($p < 0.001$)	0.217 ($p < 0.001$)

Table 2: Log-likelihood of experimental data given the next word distributions of BERT from all the focus alternative structures.

	Set	Ordering	Mean across all structures
BERT	- 466.52	- 243.66	- 355.09
BERT Static	- 890.85	- 1,462.91	- 1,176.88

References

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